Streamflow Generation in a Forested Watershed, New Zealand

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A 0.3028-ha watershed has been instrumented to monitor streamflow and subsurface flow through the soil mantle at a variety of topographic locations. The watershed is forested, with steep (35°) slopes and shallow (average 55 cm) soils on impermeable Old Man gravels. Data for a number of storms indicate that subsurface flow via 'macropores' (root channels, pipes) and seepage zones in the soil is the predominant mechanism of channel stormflow generation in storms with quickflows greater than about 1 mm. Subsurface flow from all parts of the watershed appears to contribute to stormflow even in very small storms (quickflow of the order of 3% of net precipitation). The saturated hydraulic conductivity of the soil matrix is not a limiting factor on the ability of subsurface flow to generate channel stormflow, because dye tracer experiments demonstrate that water may move through macropores (particularly root channels) at rates 2 orders of magnitude greater. However, subsurface flow from lower slope areas contributes to delayed flow; cessation of subsurface flow and streamflow after a drought period is roughly coincident in time. In the study area it appears that streamflow is at almost all times dominated by subsurface flow and that runoff from partial and variable source areas contributes significant quantities of streamflow only during the rising limb of small (less than 1 mm of quickflow) flood hydrographs.

INTRODUCTION

In recent years, considerable advances have been made in our understanding of the generation of streamflow and storm runoff, and a number of models are now available to describe mechanisms whereby precipitation appears as streamflow. Briefly, these mechanisms are (1) Hortonian overland flow, where rainfall rate exceeds the infiltration rate of the soil and the precipitation excess flows over the ground surface; (2) saturation overland flow from 'partial source areas,' in which high water tables cause saturation of small areas of a watershed, often near stream channels, and generate overland flow; (3) subsurface flow of infiltrated water moving laterally through the soil mantle toward the channel system; and (4) expansion of the channel system during storms to tap subsurface flow systems and permit overland flow from 'variable source areas.' *Dunne* [1978] and *Freeze* [1974] have provided thorough reviews of these mechanisms.

Much effort is currently going to define the situations in which each mechanism is important. It is now widely accepted that Hortonian overland flow is important primarily on unvegetated surfaces (desert environments, arable farmland, urban areas) where infiltration rates are low. Present thinking seems to emphasize the importance of stormflow generation on partial and variable source areas and to give subsurface stormflow a secondary role. *Freeze* [1974, p. 632] doubted that subsurface flow could deliver sufficient water to provide significant contributions to stormflow, citing the results of *Hewlett and Hibbert* [1967], *Weyman* [1970], and *Whipkey* [1965], which are frequently taken to support its importance. He concluded that 'only on convex hillslopes that feed deeply incised channels, and then only when the saturated hydraulic conductivities of the soil are very large, is subsurface storm flow a feasible mechanism.' More recent work by *Corbett et al.* [1975] and *Pilgrim et al.* [1978] does provide evidence for subsurface stormflow under less restrictive conditions, however.

As the partial and variable source area concepts become more widely accepted, water resource management decisions are increasingly being made in the light of their implications [*Dunne et al.*, 1975; *Engman*, 1974; *Hayward*, 1977; *Pearce and McKerchar*, 1979]. It is therefore necessary to define very closely the environments in which they are dominant and to test the predictions of theoretical studies such as those of *Freeze* [1972] against field data. The present study has monitored streamflow generation in a forested watershed to provide additional information on the relative importance of the stormflow-generating mechanisms.

STUDY AREA

Description

The study was located in one of eight experimental watersheds in Tawhai State Forest, near Reefton, New Zealand (Figure 1). The watersheds are underlain by moderately weathered, firmly compacted lower Pleistocene Old Man gravels, which are effectively impermeable. Soils are broadly classified as Blackball hill soils, shallow, strongly podzolized yellow-brown earths with a well-developed upper organic humus mantle (Figure 2). *Webster* [1977] found a substantial variation in soil depths and characteristics over very short distances (cf. Figure 3). The forest vegetation consists of undisturbed mixed beech-podocarp-hardwood forest [*Webster*, 1977] with a dense ground cover of ferns and small shrubs. Mean annual rainfall for the period 1963–1975 was 2610 mm [*Pearce et al.*, 1977], and annual runoff approximately 1500 mm.

Instrumentation

Gross precipitation is measured by a Lambrecht recording rain gauge (time resolution of 240 mm/d) just outside the main watershed (Figure 1). Throughfall through the forest canopy (net precipitation) is caught by 90 m of 10-cm-wide plastic guttering, leading to interconnected drums in which the water level is monitored by a Belfort FW1 water level recorder (time resolution of 45.5 mm/d) [*Rowe*, 1976].

Runoff from the main watershed, which has an area of 3.84 ha, is monitored by a 90° V-notch weir equipped with a Belfort FW1 water level recorder.

The hydrologic response of an 0.3028-ha subwatershed is monitored intensively by a variety of equipment (Figure 3). Streamflow in the permanent (but ephemeral) stream channel is monitored at sites C and D by 11° and 11°/90° compound V-notch weirs, respectively, each fitted with a Foxboro type

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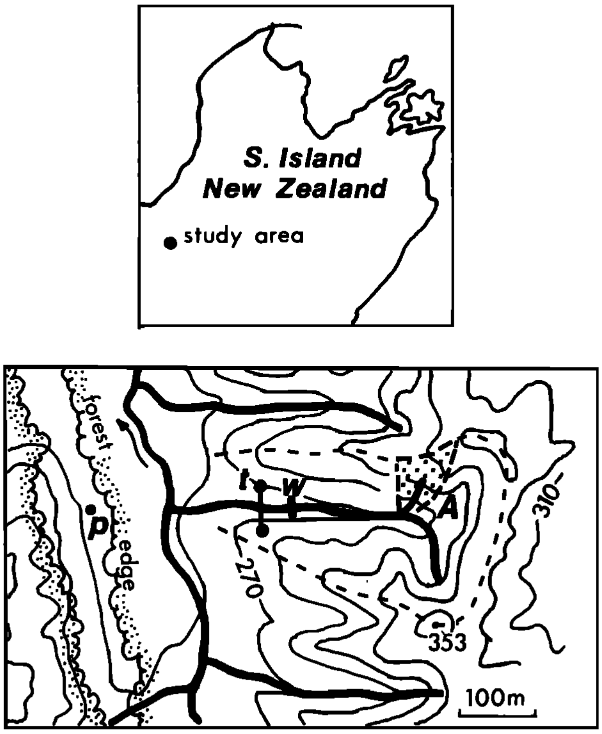

Fig. 1. Location of study area: p, rain gauge; t, throughfall measurement site; w, main weir; and A, experimental watershed (Figure 3). The contour interval is 20 m.

12R water level recorder (time resolution of 69.3 mm/d at zero flow).

Subsurface flow through the soil mantle is monitored at pits 1–6 and site A, which represent a wide range of topographic locations within the subwatershed. At each location a pit 2–3 m long was dug down through the soil into the underlying consolidated gravels, and a trough 1 m long (3.3 m at site A) was constructed at the base of the pit to intercept subsurface flow. By extending the pits beyond the collection troughs, convergence of flow lines toward the pits was minimized. Intercepted water is led from each trough to a tipping bucket; tips are recorded by an Esterline Angus eight-channel event recorder, running at 101.6 mm/hr. Discharge at a seepage site in the lower part of the watershed is also monitored by tipping bucket, and discharge from the seepage face at the head of the permanent channel by a 5$\frac{1}{2}$° V-notch weir equipped with a Foxboro 12R recorder.

At both seepage sites the collection troughs were placed at the base of preexisting vertical faces 1 m high cut into the Old Man gravels; seepage from within the soil mantle runs and drips down the face to the trough.

Data analysis

Because of equipment problems, particularly chart disengagement on the event recorder, completely reliable data for only 12 storms are presented (Table 1); fortunately, these cover a wide range of conditions. For each storm, total precipitation, net precipitation, total runoff at the main weir, and antecedent precipitation index API, defined by API = $\sum_{i=1}^{30} P_i/i$, where P_i is total gross precipitation on the ith day beforehand, were determined. For all sites, lag time T_{lag} and $T_{lag'}$, rise time T_{rise}, time to peak T_{peak}, and time to start of runoff T_q were computed (Figure 4 and Table 2). In addition, quickflow (as defined by *Hewlett and Hibbert* [1967, p. 280]) was computed for those sites (main weir, pit 1, site A, and site B) whose catchment areas could be confidently determined (Table 1). The separation line slope used by Hewlett and Hibbert (0.0055 l s^{-1} ha^{-1} hr^{-1}) was adopted for purposes of standardization and because no information is available to warrant adoption of a different value. Total runoff was taken to include quickflow and delayed flow only up to the time that quickflow ends.

It should be noted at this point that the time that runoff commences at the sites monitored by tipping buckets is not measured entirely accurately because of the time taken to fill the buckets and cause the first tip. At the very low discharge rates at the beginning of a storm, this time period may be quite long and lead to a significant overestimate of the time to start of runoff.

WATERSHED RESPONSE DURING A TYPICAL STORM EVENT

Subsurface Flow and Streamflow Hydrographs

At this stage the hydrologic response of the study watershed may be most clearly demonstrated by presenting data for a representative storm event. The storm of July 4–6, 1978, followed a prolonged dry period (API = 4.47 mm), and flow had ceased at all points within the catchment. The response to 90.6

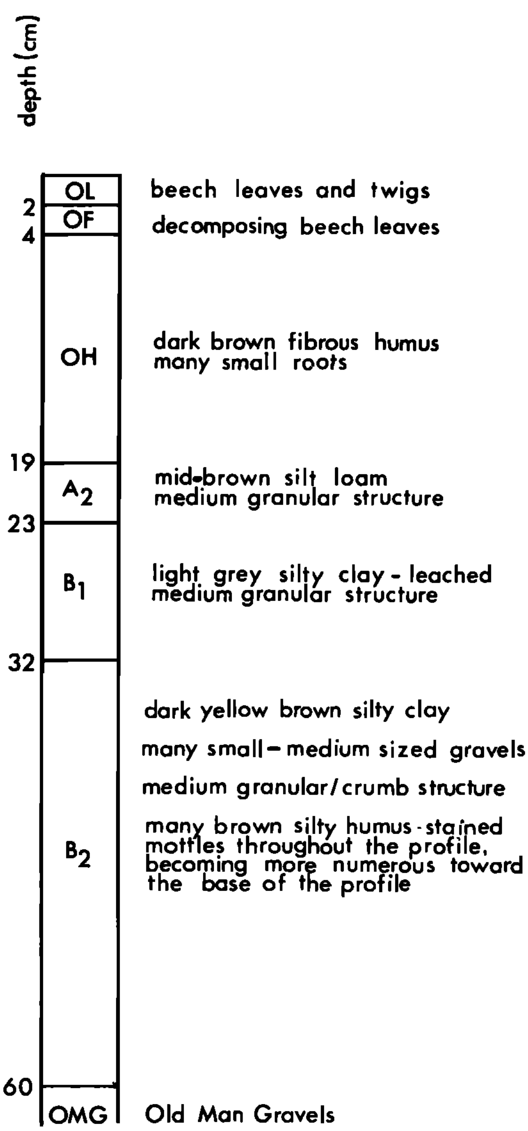

Fig. 2. Representative soil profile (sampling site 7, Table 3).

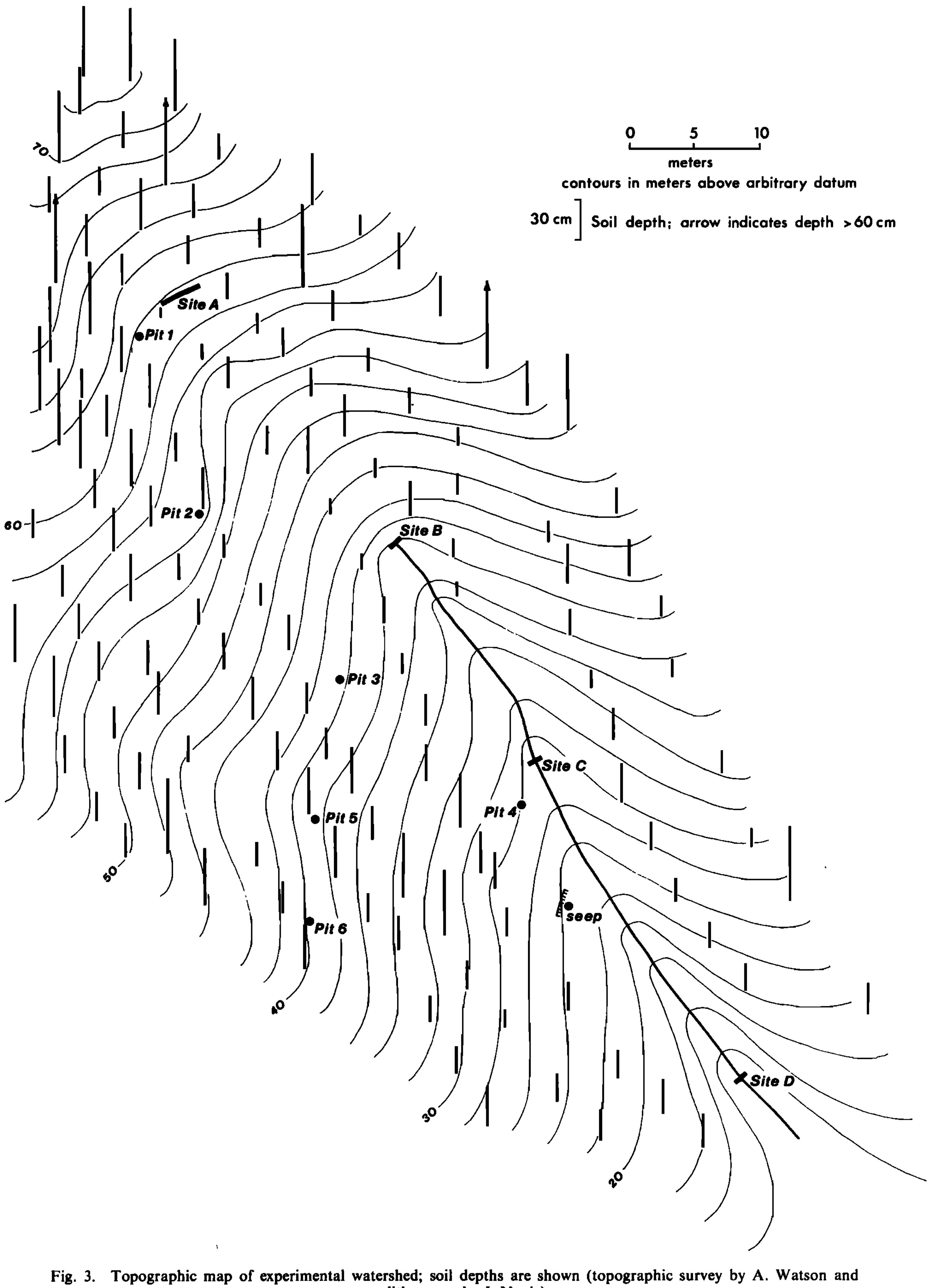

Fig. 3. Topographic map of experimental watershed; soil depths are shown (topographic survey by A. Watson and regolith survey by J. Novis).

TABLE 1. Throughflow and Streamflow During Selected Storms

Storm	Starting Date, 1978	Antecedent Precipitation Index	Total Precipitation	Net Precipitation	Total Runoff, Main Weir	Quickflow, mm				Peak Discharge, l/min												Quickflow Net Precipitation, %
						Main Weir	Pit 1	Site A	Site B	Pit 1	Pit 2	Pit 3	Pit 4	Pit 5	Pit 6	Seep	Site A	Site B	Site C	Site D	Main Weir	
1	May 6	5.58	2.6	0.40	0.07*	···†	0	0	0	0	0	0	0	0	0	0	0	0	>0.0	0.60	2.94	<<1.0
2	April 26	10.57	2.3	0.93	1.59	0.27	0	0	0.17	0	>0	0	0.013	0	0	0.096	0	0	1.44	1.44	10.2	29.0
3	April 29	6.55	7.8	5.1	0.25	0.05	0.07	0	0.01	0.0033	0	0.008	0.045	0	0	0.29	0	0.54	3.77	3.06	22.8	1.0
4	Aug. 23	26.55	7.7	5.3	1.05	0.35	0.32	0.11	0.14	0.01	0	0.008	0.044	0.031		0.47	0.125	2.40	9.00	16.2	77.4	6.6
5	April 30	11.24	8.9	9.1	1.75	1.31	1.4	0.1	1.3	0.067	0.20	0.13	0.24	0.041	0.005	1.47	0.042	2.34	16.8	30.0	200	14.4
6	July 27	41.40	18.4	14.2	9.6	5.3			7.7	0.13	0.12	0.13	0.21	0.88	0.047	1.37	0.85	7.20	16.8	33.6	332	37.5
7	May 31	12.54	23.6	17.3	2.5	1.9	2.6	0.2	2.2	0.049	0.05	0.056	0.17	0.017	>0	1.57	0.054	2.70	16.8	42.6	163	3.3
8	Aug. 24	30.00	33.5	21.4	14.7	9.8	17.1	9.7	10.7	0.17	0.16	0.12	0.24	0.75		1.59	0.90	10.2	29.4	48.6	422	45.9
										0.15	0.10	0.13	0.11	1.03		2.25	1.13	10.8	29.4	49.8	413	
9	Aug. 3	19.32	45.9	37.5	23.6	19.3	22.8	13.0	25.8	0.54	0.50	0.77	0.64	2.50		4.25	5.56	38.4	80.4	169	431	51.5
10	July 23	15.08	90.0	78.5	62.1	56.8	64.0	60.2	73.1	0.65	0.71	1.03	1.26		0.24	7.50	7.21	55.8	112	140	1830	7.20
11	Aug. 10	17.47	82.4	79.0	58.0	50.8	58.4	57.2	68.1	0.88	0.87	1.61	4.95	8.25		17.5	17.2	111.0	223	810	4040	64.3
12	July 4	4.47	90.6	80.6	46.6	43.1	45.5	45.8	59.8	1.82	2.18	2.99	7.85	19.7	1.44	23.9	34.4	115.2	257	468	4180	54.0
13	June 6	12.83	104.9	98.3	79.9	73.3				0.88	0.99	1.93	4.88	9.0	0.67	16.5	14.4	132.0	222	504	3390	74.6

Storm 8 is a double-peaked storm. Pit 6 was replaced by a time check device in July. Where there is no entry, there are no data.
*Over succeeding 24 hours.
†Slope of separation line exceeds slope of rising limb.

mm of gross precipitation (80.6 mm of net precipitation) at all the measurement sites is shown in Figures 5–8 (note the varying discharge scales). The most striking feature of the hydrographs is the close coincidence of peaks. Lag time varied between 1.0 and 1.9 hours (Table 2), with flow in pit 4, which is located in a swale, and from the seep peaking approximately 0.5 hours later than at the other sites. Although contributing areas for all sites cannot be precisely defined because of soil depth variability and contour curvature, there is a very clear increase in peak discharge and total flow volumes with distance from the watershed boundary. This indicates that flow is moving considerable distances through the soil (overland flow was observed only beside the channel between sites C and D) during the hydrograph rise. If flow at each site was derived only from the immediate locality, peak discharge and total flow volume would be independent of distance from the divide.

The subsurface flow in the subwatershed can account for the peak discharge of 69.7 l/s measured at the main weir. Within the subwatershed the peak discharge of 7.8 l/s at D can be accounted for by that at the seep (0.398 l/s), at site B (1.92 l/s), and at pit 4 (0.131 l/s per meter of valley wall). It is not possible to define precisely the contributing area at pit 4, but the pit is 44 m from the watershed divide. The mean distance from the right bank of the stream to the divide is 24 m, and from the left bank 14 m. Adjusting the pit 4 peak discharge for slope length, which is assumed to be roughly proportional to contributing area, the side slopes between B and D (48 m) would contribute 5.43 l/s, giving a total discharge at site D of 7.75 l/s, which is remarkably close to the actual measured peak discharge. The peak discharge at D (25.8 l/s/ha over the subwatershed area of 0.3028 ha) would give a peak discharge at the main weir of 99 l/s if it were supplied by the whole watershed, well in excess of the measured peak. The reasons that specific subwatershed discharge exceeds that for the whole watershed are discussed below.

Runoff Amounts

That subsurface flow can account for total streamflow throughout the storm hydrograph may be verified by computing cumulative runoff, in millimeters of water depth, for those sites for which contributing areas can be estimated (Figure 9). The cumulative runoff curves for the main weir, pit 1, and site A are quite similar; pit 1 in fact generated more runoff than the main weir during much of the storm. Site B generated substantially more runoff than the main weir during the recessions of the two hydrograph peaks, but not during the rising limb, while site D generated more runoff throughout the storm, probably because of runoff from the wet gully bottom (partial area contribution). Flow is sustained at these two locations in the subwatershed for longer and at higher levels than in the watershed as a whole; however, the close correspondence between the cumulative runoff curves for the main weir and pit 1 and site A suggest that the hydrologic performance of the whole watershed may be controlled to a large extent by subsurface flow. The sustained flow at sites B and D is presumably a function of their location in topographic hollows, which constitutes a relatively small proportion of the whole watershed.

Soil Moisture

Overland flow was not observed during the storm except in limited areas along the stream channel, so that subsurface stormflow was largely responsible for generation of the flow hydrograph. To monitor the hydrologic behavior of the soil

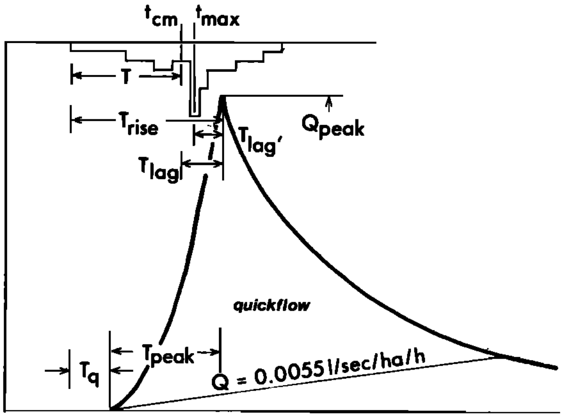

Fig. 4. Definition sketch showing hydrograph parameters.

mantle, a rectilinear slope profile stretching from valley bottom to ridge top was selected beside the throughfall measurement site, on the right bank of the main stream (t on Figure 1). A 60-cm auger was used to take soil samples in 20-cm increments from the top 60 cm of the soil at nine sites up the profile (Table 3) and at five points in time during the storm. Water contents were measured gravimetrically; because soil depths greater than 60 cm could not be sampled, total water content of the top 60 cm is presented in Figure 10, rather than percentages of the total water-holding capacity. *Webster* [1977] measured, using a laboratory permeameter, saturated water contents by volume of the mineral soil in the study area between 48% and 69% (mean, 57.6%; standard deviation s = 6.4%) and on the selected profile between 46% (28 cm of water in 60 cm of soil) at ridge top and over 70% (42 cm of water in 60 cm of soil) on the lower slopes.

Sites 1, 2, and 4 rapidly attained saturation throughout their depths, although overland flow was never observed. Free water rapidly appeared in the auger holes during sampling. Sites 1 and 2 are lower slope sites, adjacent to the valley floor, and site 4 is immediately upslope of an area of unusually thin soil (cf. site 3, Table 3). The midslope sites 3, 5, and 6 became saturated in the 20- to 40- and 40- to 60-cm depth increments; again, free water rapidly appeared in the auger holes. The upper slope sites 7, 8, and 9 approached saturation only in the 40- to 60-cm depth increment.

Samples were unfortunately not obtained before the start of the storm, but J. Webster (personal communication, 1978) considers that soil water content in the study area rarely declines below 10% of saturated water content. There was a slight tendency for water contents to decline during the hydrograph recession period; this was more marked for the upper slope sites. Broadly, the data suggest that there was rapid movement of water vertically in the soil profile and in the downslope direction, such that a saturated wedge, whose shape was modified by the shape of the bedrock surface, built up in the soil mantle. The wedge almost intersected the ground surface at the foot of the slope and above the zone of thin soil at site 4 and tapered off in the upslope direction, becoming restricted to only a thin layer above the bedrock surface at the top of the slope.

TABLE 2. Hydrograph Parameters for Selected Storms

	Pit							Site				Main
	1	2	3	4	5	6	Seep	A	B	C	D	Weir
					April 29, 1978; Qf = 1.0%; T = 6.1							
T_{lag}	4.6		4.6	3.6			3.6		2.4	2.4	1.6	3.1
$T_{lag'}$	4.2		4.2	3.2			3.2		2.0	2.0	1.2	2.7
T_{rise}	10.7		10.7	9.7			9.7		8.5	8.5	7.7	9.2
T_{peak}	1.5		0.6	5.4			5.4		1.5	1.5	1.7	3.0
T_{start}	9.2		10.1	4.3			4.3		7.0	7.0	6.0	6.2
					April 30, 1978; Qf = 14.4; T = 1.5							
T_{lag}	2.9	3.5	2.9	2.9	13.0	2.9	2.9	3.2	3.8	1.5	1.8	1.8
$T_{lag'}$	2.9	3.5	2.9	2.9	13.0	2.9	2.9	3.2	3.8	1.5	1.8	1.8
T_{rise}	4.4	5.0	4.4	4.4	14.5	4.4	4.4	4.7	5.3	3.0	3.3	3.3
T_{peak}	1.4	0.8	0.9	3.4	5.8	1.0	3.4	0.4	4.3	1.7	2.1	2.3
T_{start}	3.0	4.2	3.5	1.0	8.7	3.4	1.0	4.3	1.0	1.3	1.2	1.1
					July 27, 1978; Qf = 37.4%; T = 2.9							
T_{lag}	5.8	5.8	5.7	5.8	7.2	6.9	5.4	6.9	5.1	5.1	4.6	4.6
$T_{lag'}$	6.4	6.4	6.3	6.4	7.8	7.5	6.0	7.5	5.7	5.7	5.2	5.2
T_{rise}	8.7	8.7	8.6	8.7	10.1	9.8	8.3	9.8	8.0	8.0	7.5	7.5
T_{peak}	7.7	5.9	6.1	8.1	6.8	7.4	7.5	6.4	7.5	7.7	7.4	8.0
T_{start}	1.0	2.8	2.5	0.6	3.3	2.4	0.8	3.4	0.5	0.3	0.1	0
					July 4, 1978; Qf = 54.0%; T = 7.2							
T_{lag}	1.3	1.3	1.0	1.9	1.6	1.0	1.9	1.3	1.5	1.3	1.5	1.3
$T_{lag'}$	2.0	2.0	1.7	2.6	2.3	1.7	2.6	2.0	2.2	2.0	2.2	2.0
T_{rise}	8.5	8.5	8.2	9.1	8.8	8.2	9.1	8.5	8.7	8.5	8.7	8.5
T_{peak}	4.0	3.2	2.9	5.6	3.5	2.5	6.5	3.0	4.5	5.3	6.5	6.3
T_{start}	4.5	5.3	5.3	3.5	5.3	5.7	2.6	5.5	4.2	3.2	2.2	2.2
					June 6, 1978; Qf = 74.6; T = 16.3							
T_{lag}	2.0	3.1	2.5	3.0	2.5	2.5	3.0	2.5	0.2	0.2	0.4	1.7
$T_{lag'}$	2.8	3.9	3.3	3.8	3.3	3.3	3.8	3.3	1.0	1.0	1.2	2.5
T_{rise}	18.3	19.4	18.8	19.3	18.8	18.8	19.3	18.8	16.5	16.5	16.7	18.0
T_{peak}	12.1	9.8	11.9	13.8	5.7	6.1	14.1	7.3	18.0	18.0	18.2	18.7
T_{start}	6.2	9.6	6.9	5.5	13.1	12.7	5.2	11.5	−1.5	−1.5	−1.5	−0.7

Where there is no entry, there is no flow.

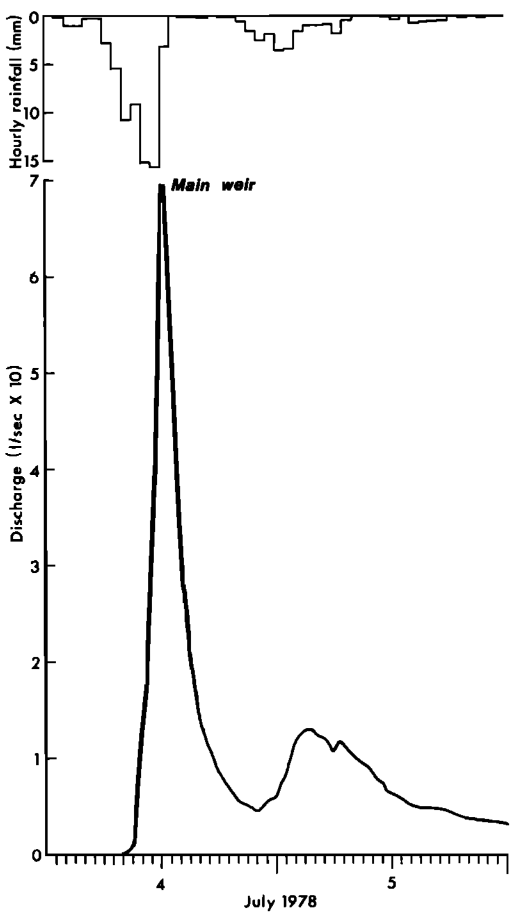

Fig. 5. Streamflow main weir, July 4–5, 1978.

MOVEMENT OF WATER THROUGH THE SOIL

Figure 11 demonstrates the extreme sensitivity and rapidity with which the subsurface flow measurement sites respond to variations in precipitation intensity; lag times ($T_{lag'}$) are, as in the July 4–6 storm, about 2 hours.

Webster [1977] measured saturated hydraulic conductivities of the organic layer and mineral soil in the study area; mean values of 0.14 cm/s (s = 0.03 cm/s) and 0.007 cm/s (s = 0.001 cm/s), respectively, were obtained. For the representative profile in Figure 2, precipitated water transmitted vertically through the soil matrix at the saturated flow rate would require $(19/0.14) + (41/0.007) = 5993$ s, or 100 min, to reach the bedrock surface. With the wide range in soil depths in the study subwatershed (Figure 3), there would be a wide variation in this time requirement, but water moving at saturated flow rates vertically down through the soil profile then along a seepage zone at the base of the B horizon could cause a rapid response at the subsurface flow measurement sites.

However, because subsurface flow discharges increase in a downslope direction (Figures 6–8), it appears that there is downslope movement of water over distances of up to several tens of meters during the rising limb of the storm hydrograph; this could not be accomplished by saturated flow through the mineral soil. Pits dug in the study watershed for soil profile description and for installation of subsurface flow interception troughs displayed points of concentrated seepage, usually at the base of the B horizon, at which high rates of outflow were observed during storms. At one site on the 28-m contour between pit 5 and the seep (Figure 3), two pipes were discovered at the base of the B horizon through which water gushed at maximum rates of the order of 20 l/s. It is inferred that this water appeared at the seep.

Aubertin [1971], *Jones* [1978], and other workers have demonstrated that water may move through the soil mantle via 'macropores' at rates considerably greater than the saturated hydraulic conductivity of the soil, and a number of trials were carried out in the present study to evaluate the significance of this mechanism. The trials were conducted at the end of a storm that delivered 34 mm of rainfall in 48 hours. Dye was applied to the soil surface through a 15-cm-diameter cylinder; the cylinder was pressed into the humus layer to prevent leakage around the side. A mean depth of dye of 2 cm was maintained during the period of application; the dye soaked rapidly into the humus, so that a positive pressure head existed for only 2–4 min, and then only beneath the cylinder. Conditions were clearly artificial, but except in the immediate vicinity of the cylinder it is considered that movement of the dye was similar to that of water applied by an intense rainfall event.

Trial 1. Six liters of sodium dichromate was applied 1 m upslope from pit 6. The first traces of dye appeared through root holes in the B horizon 5.25 min after start of application. Depth of first appearance was 20.3 cm, well above the base of the B horizon. Dye velocity was 0.38 cm/s. Dye eventually seeped from the base of the B horizon, with a lateral spread across the face of the pit of 90 cm. No seepage from the top of the A horizon was observed. Maximum outflow rate occurred 12.5 min after start of application, at a rate of 0.0019 l/s; at least 3 l of dye flowed from the pit. Dye was still seeping from the pit 5 hours after the start of the trial.

Trial 2. Rhodamine B dye was injected 0.62 m upslope from pit 6, to the right of the trial 1 location. The first traces of dye appeared 1.5 min after first application at 15-cm depth, just below the base of the A_2 horizon; after 2 min a row of dye seeps, presumably at root channels, at 26-cm depth was visible. Maximum velocity was 0.81 cm/s. Dye also appeared at the base of a large root at the bottom of the humus layer; otherwise, no flow at the top of the mineral soil was observed. Dye eventually seeped out at a maximum depth of 35 cm, with a horizontal spread of 1 m.

Trial 3. Six liters of rhodamine B was applied 1 m upslope from pit 5 (depth, 1.6 m). Sodium dichromate was added after 20.5 min, at the same site. Rhodamine B was never observed in the pit, although clear fluid did percolate from a seep at the base of the B horizon. Sodium dichromate became visible from this seep 23 min after the start of its injection and after 30.5 min from smaller seeps at the base of the B horizon. The density of the dye color increased with time, the fluid initially being almost clear, probably because of adsorption of the dye by the soil. Dye appeared nowhere else in the profile. Maximum velocity from point of application to seepage point was 0.17 cm/s.

Trial 4. A new pit 90 cm deep was dug near sampling site 8 on the soil moisture sampling transect. At a point 4 m upslope from the pit applications of water were made as follows.

Elapsed time t, min	Volume, l
0–4	24
25–28	23
45–50	32

The first seep into the pit occurred at t = 54 min 30 s in the A_2 horizon, just below the base of the humus, and a second at t = 55 min 22 s. These two seeps appeared to be old root channels. Smaller seeps started to appear in the A_2 horizon

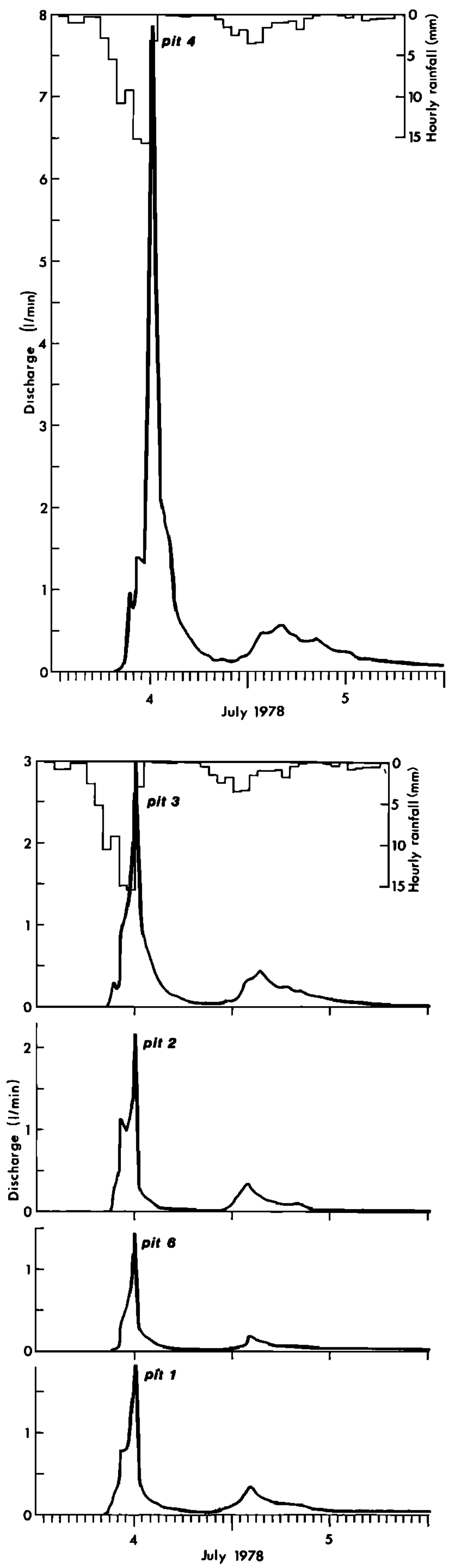

Fig. 6. Subsurface flow at side slope sites, July 4–5, 1978.

after 59 min; lateral spread from the injection point was 1 m. It is inferred that the first two applications of water (equivalent to approximately 15 mm of water over the area between the injection point and the pit) were completely soaked up by the soil and that the seepage observed came from the third application. A velocity of travel of the order of 0.7 cm/s is indicated if this inference is correct. Peak outflow occurred at t = 70 min 15 s.

At t = 105 min to 107 min 30s, 23 l of water were injected at a point 3 m upslope from the pit. A significant increase in outflow from the main seeps was observed at t = 108 min 15 s. A travel velocity of 0.54 cm/s is indicated. Peak flow occurred at t = 111 min.

At t = 145 min to 147 min 30 s, 10 l of water dyed with lissamine green dye was injected 2 m upslope from the pit. First seepage of dyed water was observed at t = 148 min; a travel velocity of 1.11 cm/s is indicated. Peak outflow occurred at t = 150 min. No seepage was observed at any point in the profile other than in the A_2 horizon; the B horizon has no observable root channels, unlike the other sites at which dye-tracing experiments were conducted. However, seepage from the base of the profile, at the soil-bedrock interface, was observed before the trial was started, and this appeared to be delayed seepage from a similar trial carried out in the pit 18 hours previously.

Additional tracer experiments were carried out toward the end of a storm which delivered 82 mm of precipitation in 26 hours. In each case, 1 l of rhodamine B dye was simply sprinkled onto the soil surface 1 m upslope from the pit; the dye soaked immediately into the humus layer.

Trial 5. The trial was conducted at pit 6. Outflow was first observed after 82 s in the B horizon, giving a maximum travel velocity of 1.2 cm/s, in comparison with a maximum velocity of 0.38 cm/s in trial 1 at the same site. Dye soon appeared at the row of seeps at 20- to 25-cm depth (cf. trial 2).

Trial 6. The trial was conducted at the trial 4 site. Outflow was observed after 60 s at the same seepage points observed in trial 4. Maximum travel velocity was 1.7 cm/s, in comparison with 0.54 cm/s in trial 4.

Trials 7 and 8. Similar trials at pit 2 and site A gave maximum travel velocities of 1.4 and 2.1 cm/s; at these sites, dye appeared at the base of the humus layer.

A search was made for locations at which overland flow velocities could also be measured. Four sites, three of which were in swales, were found at which water was moving downslope over the forest floor as unchanneled flow (although there was a tendency for flow to concentrate into an anastomosing system of small rivulets). Travel velocities measured by dye injection averaged 4.25 cm/s, in comparison with the mean of 1.6 cm/s for the four subsurface flow tests.

Maximum dye travel velocities in the subsurface flow trials were up to 300 times greater than the saturated hydraulic conductivity of 0.007 cm/s for mineral soil. At some locations, much of the fluid applied at the surface reappeared downslope at the base of the humus layer; at these sites a paucity of roots and root channels in the A and B horizons was observable. At other sites, much of the fluid reappeared from seeps, usually associated with root channels, or at the base of the B horizon, above the impermeable bedrock surface. At these latter sites, there appeared to be a greater density of roots and root channels in the mineral soil.

The dye-tracing experiments and other storm period observations permit a qualitative discussion of water movement through the soil mantle. Because of the considerable variability in soil and root network characteristics, topography, an-

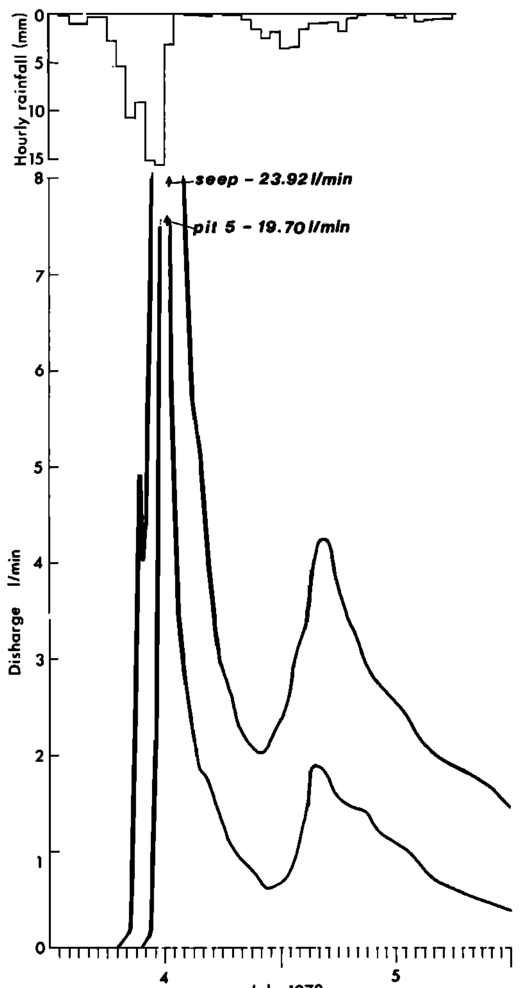

Fig. 7. Subsurface flow at swale and seep sites, July 4–5, 1978.

tecedent moisture conditions, and storm characteristics it is not feasible to apportion quantitatively amounts of water moving along different pathways, although further work is currently in progress to provide more detailed information. Water reaching the surface of the forest floor rapidly enters the organic layer, moving in a predominantly vertical direction through the highly porous humus (values for mean total porosity and macroporosity of 15 samples are 86% and 39% by volume, respectively [*Webster,* 1977]. Macroporosity is defined here as the percentage of the soil volume drained at a tension of 50 cm of water for 24 hours. It is distinct from the term macropore used elsewhere in the text, which refers qualitatively to such features as root channels, animal burrows, pipes, or cracks in the soil through which water may move. These will contribute to the macroporosity as defined above, but only partially.) A variable proportion of this water is held by the organic layer, depending on antecedent moisture conditions, precipitation intensity, and depth of the organic layer; J. Webster (personal communication, 1978) estimates that on average there would be 8 mm of unsatisfied water storage capacity in this layer. The remainder reaches the interface between the organic and mineral soils, at which the saturated hydraulic conductivity and porosity decline significantly.

Although the saturated hydraulic conductivity value of 0.007 cm/s for the mineral soil implies that up to 25 cm/hr of water could pass directly into the mineral soil, a large proportion in fact runs downslope above the surface of the A horizon and within its top 1–2 cm, which is more loosely aggregated and more porous than the rest of the mineral soil. The delayed seepage observed in trials 1 and 4 and apparent in subsurface flow hydrographs indicates that water enters and moves through the mineral soil at rates in line with the saturated hydraulic conductivity, but the rapidity of flow above the interface presumably prevents infiltration of all the water before it reaches the stream channels. A distinct saturated zone 1–2 cm deep in the base of the organic layer was frequently observed during storm conditions, and it is inferred that water moves downslope through this saturated, highly porous layer in a manner intermediate between free surface flow and flow through a porous medium (cf. the relative magnitudes of saturated hydraulic conductivity, subsurface flow velocities, and overland flow velocities).

Where roots, decayed root channels, and other macropores with diameters greater than about 3 mm (at which capillary forces become unimportant) provide pathways across the interface between the organic and mineral horizons, water moves rapidly into the mineral soil under the force of gravity and eventually reaches the interface between the B horizon and the impermeable Old Man gravels. A seepage zone has in most places been observed above this interface, and in some locations, pipes exist. The shallow, indefinite nature of the seepage zone has prevented measurement of characteristics such as porosity and hydraulic conductivity, but it seems probable that eluviation has increased hydraulic conductivities there.

That water may move rapidly along macropores and seepage zones rather than be drawn into the soil matrix is considered to be a function of the soil moisture conditions prevailing in the study area. As has already been noted, soil water contents rarely decline more than 10% below saturation. Hence

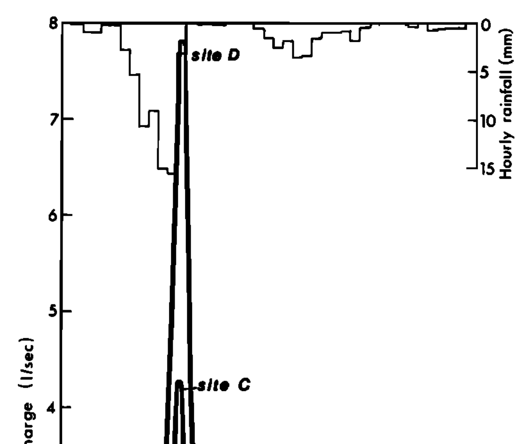

Fig. 8. Subsurface flow and streamflow up axis of experimental watershed, July 4–5, 1978.

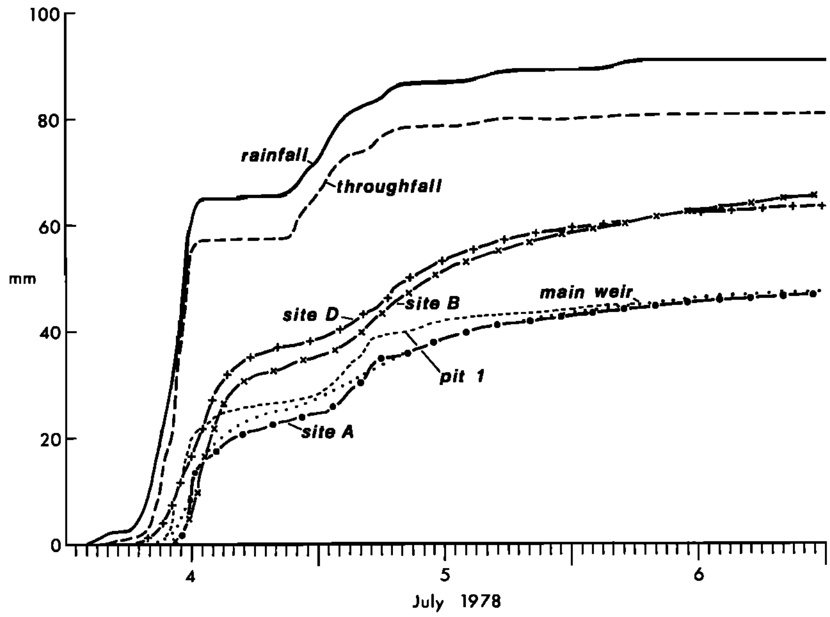

Fig. 9. Cumulative runoff at selected sites, July 4–6, 1978.

the mineral soil, especially below about 20-cm depth, may become saturated at an early stage of a storm and permit flow along macropores without losses to the soil matrix. Even where the soil matrix is unsaturated, water may not be drawn into the matrix rapidly enough to prevent flow reaching the saturated layers or the stream channel.

The processes of water movement through the soil in the study area require much more investigation, but even then they may prove so variable in response to variations in soil characteristics as to prevent precise description. Nevertheless, the observations reported herein demonstrate that subsurface flow through macropores and along discontinuities in the soil profile is capable of contributing to storm period streamflow and suggest that the translatory flow mechanism discussed by *Hewlett and Hibbert* [1967] may not be important in the study area.

CONTRIBUTION OF SUBSURFACE FLOW TO CHANNEL QUICKFLOW

The recent emphasis on the partial and variable source area concepts of streamflow generation has led to the view that there is a close relationship between the percentage of precipitation appearing as quickflow and the percentage of the total catchment area contributing to stormflow. In other words, as the percentage of precipitation appearing as quickflow increases, it is assumed that an increasing proportion of the watershed is directly contributing to storm runoff.

TABLE 3. Soil Moisture Sampling Site Characteristics

Sampling Site	Distance From Base of Slope, m	Depth of Organic Horizon, cm	Depth of Whole Profile, cm
1	0	3.0	>100
2	11.0	5.5	91.5
3	19.5	21.0	45.0
4	30.0	4.5	75.5
5	38.4	12.5	81.5
6	50.3	19.0	60.0
8	66.0	18.0	63.0
9	76.0 (ridge crest)	45.0	59.0

For the period December 1974 to February 1977, *Pearce and McKerchar* [1979] derived figures for precipitation, total runoff, and quickflow in the study watershed:

gross precipitation	5573 mm;
net precipitation	4403 mm;
total runoff	3275 mm;
quickflow	2201 mm.

Quickflow was 50% of net precipitation and 67% of total runoff. However, in 74 individual storms producing at least 0.5 mm of quickflow, quickflow as a proportion of net rainfall ranged from 4.3% to 99%, and Pearce and McKerchar concluded that 'the area of the catchment contributing to stormflow ranges from less than 5% to practically the whole catchment during some large storms with wet antecedent conditions.' They demonstrated that overland flow due to an excess of precipitation rate over infiltration rate or to runoff from permanently saturated wetlands (about 5% of total watershed area) could not alone generate storm runoff and sug-

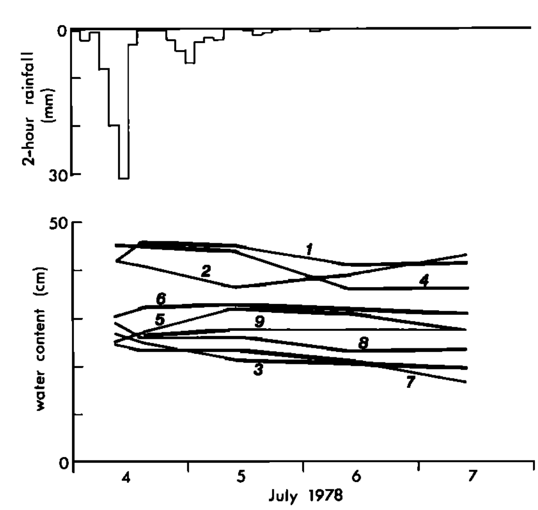

Fig. 10. Water contents of top 60 cm of soil, hillslope transect in experimental watershed, July 4–7, 1978. Note that sample site 3 has a soil depth of only 45 cm.

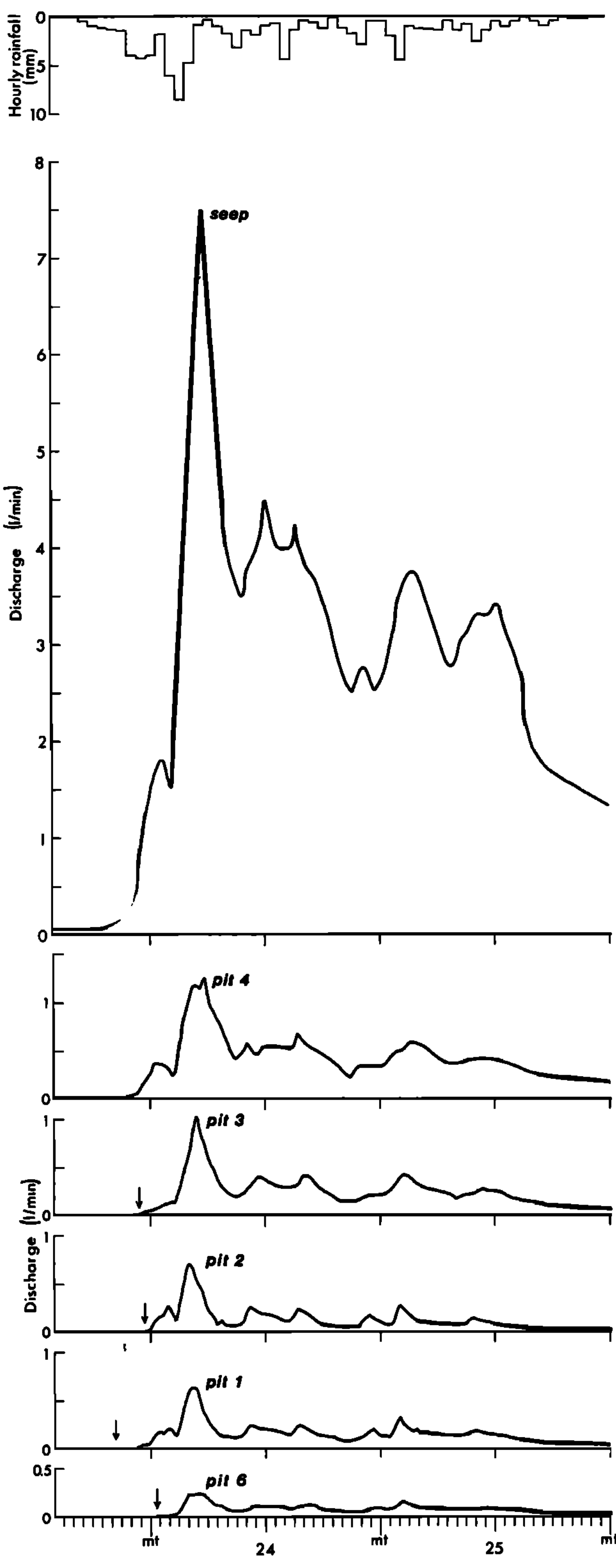

Fig. 11. Subsurface flow at side slope and seep sites, July 23–25, 1978.

gested that in events yielding more than 5 mm of quickflow the bulk of the storm runoff can be accounted for only by high rates of subsurface flow.

The data presented above for the July 4–6 storm (43 mm of quickflow) are in accord with Pearce and McKerchar's conclusions, but information for other storms indicate that they have underestimated the importance of subsurface flow in smaller events.

Thirteen storms have been ranked in Table 1 in terms of net precipitation; data are presented for peak discharges at all study sites and for quickflow at the sites whose contributing areas may be defined. Subsurface flow occurs in very small storm events in quantities sufficient to contribute a significant proportion of the runoff from the main weir. For example, in storm 7, quickflow at the main weir was 1.9 mm (3.3% of the net precipitation of 17.3 mm), at pit 1 it was 2.6 mm, and at site B 2.2 mm. The low figure for quickflow of 0.2 mm at site A is probably because the greater than average soil depths in its contributing area store a larger proportion of net precipitation in the early stages of a precipitation event; site A produces smaller quantities of quickflow than the main weir up to about 10 mm of quickflow. The large difference between the quickflows for pit 1 and site A, which are only 4 m apart, is indicative of the influence of slope shape and soil depth. Pit 1 is on a planar slope with a shallow (20 cm deep) soil, whereas the catchment area of site A is bowl shaped, and the soil at the site is 45 cm deep.

Subsurface flow does not occur throughout the study watershed until quickflow at the main weir exceeds 1.31 mm (storm 5, Table 1; net precipitation, 9.1 mm), but some sites respond when as little as 0.05 mm of quickflow is produced at the main weir (storm 3, Table 1; net precipitation, 5.1 mm). A plot of quickflow at the main weir against quickflow at the subsurface flow sites (Figure 12) indicates that quickflow at these sites is equal to or greater than that at the main weir at quickflows above 1 mm. Even below 1 mm, measurable quickflow at these sites may occur, but it tends to zero below this value, depending on antecedent precipitation.

The regression relationships between peak specific discharges in liters per second per hectare at the main weir and at pit 1 and sites A, B, and D for the storms in Table 1 (zero values omitted) are

$$\begin{aligned} Q_m &= 0.799Q_1^{0.82} & r^2 &= 0.92 \\ Q_m &= 1.98Q_A^{0.54} & r^2 &= 0.83 \\ Q_m &= 0.832Q_B^{0.87} & r^2 &= 0.93 \\ Q_m &= 0.532Q_D^{0.98} & r^2 &= 0.95 \end{aligned} \tag{1}$$

Equality is attained at

$$\begin{aligned} Q_m &= Q_1 = 0.3 \text{ l/s/ha } (1.15 \text{ l/s}) \\ Q_m &= Q_A = 1.42 \text{ l/s/ha } (5.45 \text{ l/s}) \\ Q_m &= Q_B = 0.25 \text{ l/s/ha } (0.96 \text{ l/s}) \end{aligned} \tag{2}$$

(where the values in parentheses are those at the main weir) and not at all for site D. Equations (1) and (2) indicate that in storms generating more than about 1 mm of quickflow, peak specific throughflow discharges exceed peak specific discharges in the channel and increase rather more quickly with storm size. This is in line with interpretation of Figure 12. The relationship for site A is strongly influenced by the lack of runoff during small storm events, which is probably caused by loss of precipitation to soil moisture storage. Peak specific discharge in the subwatershed above site D is double that for the main watershed as a whole, irrespective of storm size; precipitation into the channel and runoff from saturated areas along the channel may account for the difference in smaller storms but would rapidly become relatively unimportant with increasing storm size.

The study subwatershed is apparently not entirely representative of the whole watershed above the main weir, in that

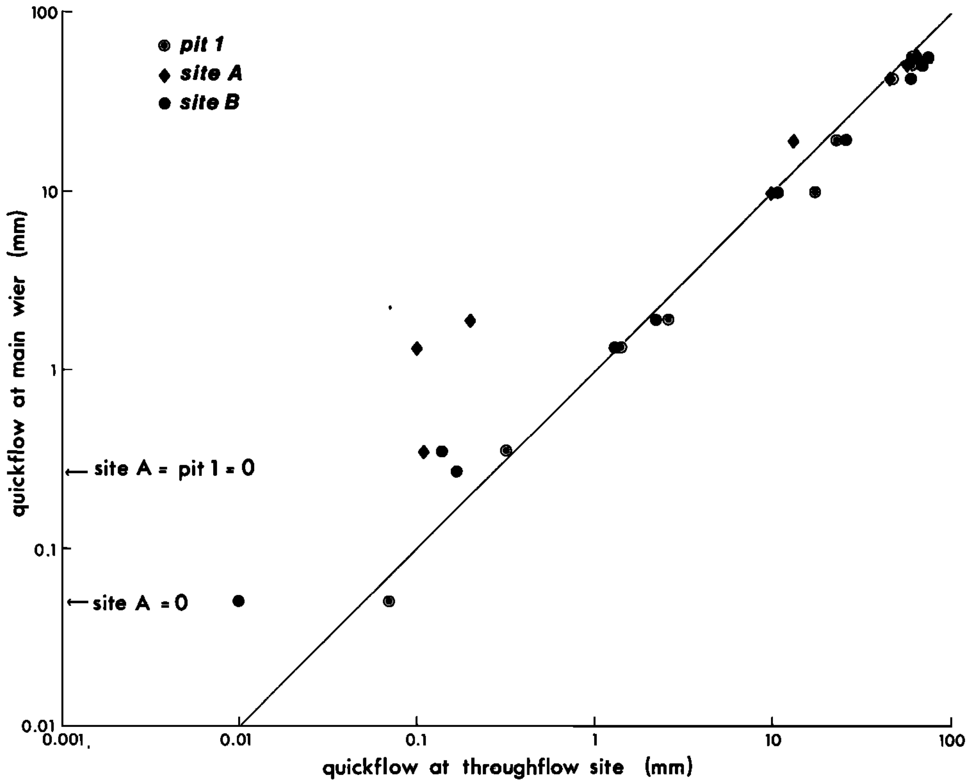

Fig. 12. Relation between quickflow at main weir site and quickflow at selected subsurface flow measurement sites.

subwatershed storm period specific discharges generally exceed those for the whole watershed. The subwatershed is a topographic hollow and hence might be expected to concentrate runoff to a greater extent than the side slopes and spurs that predominate in the main watershed. Furthermore, soils appear to be shallower than average.

SUBSURFACE FLOW CONTRIBUTIONS TO LOW STREAMFLOW

Quickflow in the storm of July 4–6, 1978, terminated at 1500 hours, July 6, at a discharge of 1.4 l/s at the main weir. Discharges at the throughflow sites are presented in Table 4; the final bucket tips in pits 2 and 6 were at 1942 and 1509 hours, respectively. Throughflow from site B, the seep, and pit 4 (adjusted as before) can account for discharge at site D, but specific discharge at site D (0.27 l/s/ha) is less than at the main weir (0.36 l/s/ha). This is consistent with the proportionately larger contribution to stormflow by the study subwatershed and suggests that other areas in the main watershed contribute proportionately less to stormflow and more to delayed flow. (Some areas with deep soils and a denser than average vegetation cover will presumably contribute less to both stormflow and delayed flow.)

Other storms amplify the relative importance of subsurface flow with respect to channel quickflow and delayed flow. For example, the storm event of July 4–6, discussed earlier, was preceded by 24 days without rain. The preceding storm, on June 7–9, produced 116.7 mm of precipitation; the precipitation center of mass was at 0400 hours on June 8. Quickflow at the main weir ceased at 1900 hours on June 10, and flow ceased altogether at 1200 hours on July 1. Table 5 shows the dates and times of the last bucket tip at each subsurface flow site; periods of flow may be underestimated because flow insufficient to fill the buckets may have continued. All sites continued to discharge after the end of channel quickflow, but the three upper slope sites did so for considerably shorter periods than the lower slope and swale sites. The time at which flow ceased at the lower slope sites coincides roughly with the

TABLE 4. Subsurface flow Discharges at End of Channel Quickflow Period, 1500 Hours, July 6, 1978

Site	Discharge at Subsurface Flow Site, l/min
Pit 1	0.009
Pit 2	0.001
Pit 3	0.013
Pit 4	0.048
Pit 5	0.189
Pit 6	0.02
Seep	0.79
Site A	0.39
Site B	3.6

TABLE 5. Periods of Flow Following Storm of June 7–9

Site	Last Bucket Tip	Period of Flow From Storm Center of Mass, hours
Pit 1	1409 hours, June 14	154
Pit 2	2315 hours, June 10	67
Pit 3	0157 hours, June 15	166
Pit 4	1749 hours, June 30	542
Pit 5	1425 hours, June 29	514
Pit 6	bucket faulty	
Seep	0214 hours, July 3	594
Site A	1106 hours, June 23	367
Site B	2400 hours, July 1	572

time that flow ceased at the main weir; subsurface seepage past the weir may explain why flow there ceased earlier.

Taken together, the data indicate that not only is subsurface flow important in generating stormflow in the channel system, but it also accounts for base flow, as was suggested by *Hewlett* [1961]. It is inferred that the contribution to stormflow is via macropores in the soil mantle and that slow release by saturated and unsaturated flow of the moisture that enters the soil matrix contributes to base flow discharge.

CONCLUSIONS

It is important to define the situations in which each streamflow-generating mechanism is important, so that implications for management can be confidently drawn. The present detailed study partially confirms the conclusions of *Freeze* [1972, 1974] and *Pearce and McKerchar* [1979] regarding the relative significance of subsurface stormflow as a contributor to streamflow but considerably extends their results.

In the study area it appears that direct runoff from precipitation on partial and variable source areas could be dominant only in those storms generating less than 1 mm of quickflow. In such small storms, however, overland flow is unlikely, and precipitation into the channel would be of more significance. Otherwise, subsurface flow is capable of accounting for the bulk of the channel quickflow. *Freeze* [1972, p. 1282] considered that a threshold value for saturated hydraulic conductivity of the order of 0.002 cm/s is necessary for subsurface stormflow to be significant, but in a soil that contains root channels, pipes, and seepage zones, saturated hydraulic conductivity of the soil matrix is not a limiting factor. Flow of dye tracer through macropores in the soil was observed at rates up to 3 orders of magnitude greater, and the sensitive and rapid response of subsurface flow to variations in precipitation suggests that flow through macropores rather than through the soil matrix contributes to channel stormflow. This flow was of 'new' water, and no evidence for translatory flow was observed. In addition, slow drainage of water from the soil mantle by unsaturated flow (and possibly saturated flow in a shallow layer above the soil-bedrock interface) appears to account for delayed flow throughout the hydrograph recession.

The notion that the proportion of precipitation appearing as quickflow is a good indicator of the total proportion of the watershed contributing to quickflow does not appear to apply to the study area. Subsurface flow was observed virtually throughout the study watershed in storms with as little as 3.3% of net precipitation appearing as quickflow, and subsurface flow from all parts of the subwatershed probably contributes to streamflow in storms with quickflows greater than 1 mm. The channel in the subwatershed is admittedly incised, so that the maximum area of near-channel wetlands is fairly restricted, but this does not fundamentally affect the conclusion that subsurface flow is the dominant mechanism of both stormflow and base flow generation in the study area.

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REFERENCES

Aubertin, G. M., Nature and extent of macropores in forest soils and their influence on subsurface water movement, *U.S. Forest Serv. Res. Pap., NE-192*, 33 pp, 1971.

Corbett, E. S., W. E. Sopper, H. W. Lull, Watershed response to partial area applications of simulated rainfall, *Int. Ass. Hydrol. Sci. Publ., 117*, 63–71, 1975.

Dunne, T., Integration of hillslope flow processes, in *Hillslope Hydrology*, edited by M. J. Kirkby, chap. 7, John Wiley, New York, 1978.

Dunne, T., T. R. Moore, and C. H. Taylor, Recognition and prediction of runoff-producing zones in humid regions, *Hydrol. Sci. Bull., 20*, 305–327, 1975.

Engman, E. T., Partial area hydrology and its application to water resources, *Water Res. Bull., 10*, 512–521, 1974.

Freeze, R. A., Role of subsurface flow in generating surface runoff, 2, Upstream source areas, *Water Resour. Res., 8*, 1272–1283, 1972.

Freeze, R. A., Streamflow generation, *Rev. Geophys. Space Phys., 12*, 627–647, 1974.

Hayward, J. A., Hydrology of a mountain catchment, and its implications for management, in *Proceedings of the 3rd Soil and Plant Water Symposium*, pp. 126–136, New Zealand Department of Scientific and Industrial Research, Wellington, 1977.

Hewlett, J. D., Soil moisture as a source of baseflow from steep mountain watersheds, *U.S. Forest Serv. Res. Pap., SE132*, 11 pp., 1961.

Hewlett, J. D., and Hibbert, A. R., Factors affecting the response of small watersheds to precipitation in humid areas, in *Forest Hydrology*, edited by W. E. Sopper and H. W. Lull, pp. 275–290, Pergamon, New York, 1967.

Jones, J. A. A., Soil pipe networks: Distribution and discharge, *Cambria, 5*, 1–21, 1978.

Pearce, A. J., and A. I. McKerchar, Upstream generation of storm runoff, in *Physical Hydrology—New Zealand Experience*, edited by D. L. Murray and P. Ackroyd, New Zealand Hydrological Society, Wellington, in press, 1979.

Pearce, A. J., C. L. O'Loughlin, and L. K. Rowe, Hydrologic regime of small, undisturbed beech forest catchments, North Westland, in *Proceedings of the 3rd Soil and Plant Water Symposium*, pp. 150–158, New Zealand Department of Scientific and Industrial Research, Wellington, 1977.

Pilgrim, D. H., D. D. Huff, and T. D. Steele, A field evaluation of subsurface and surface runoff, II, Runoff processes, *J. Hydrol., 38*, 319–342, 1978.

Rowe, L. K., Preliminary results from a rainfall interception study under a podocarp/beech/hardwood forest in North Westland, New Zealand, in *Proceedings of the New Zealand Hydrological Society Symposium*, pp. 151–164, New Zealand Hydrological Society, Wellington, New Zealand, 1976.

Webster, J., The hydrologic properties of the forest floor under beech/podocarp/hardwood forest, North Westland, M.Ag.Sci. thesis, 77 pp., Univ. of Canterbury, Canterbury, New Zealand, 1977.

Weyman, D. R., Throughflow on hillslopes and its relation to the stream hydrograph, *Bull. Int. Ass. Sci. Hydrol., 15*, 25–33, 1970.

Whipkey, R. Z., Subsurface stormflow from forested slopes, *Bull. Int. Ass. Sci. Hydrol., 10*, 74–85, 1965.

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